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112006202 helped in discussing about ambiguity of questions (especially Q1 and Q2)

I have done this assignment without the use of any AI agent or tool

Week 2 Report

Reusable function that does the plot for Q1:

```r
plot_distribution <- function(data, title) {
  # plot the density function of data
  plot(density(data), col="blue", lwd=2, main = title)

  # add vertical lines showing mean (solid) and median (dashed)
  abline(v = mean(data))
  abline(v = median(data), lty = "dashed")
}
```

Code 0. Reusable function for Q1's plot.

Q1.(a) Distribution 2: a combined dataset (n=800) that is negatively skewed by changing the mean and standard deviation of d1, d2, and d3.

```r
d1 <- rnorm(n=500, mean=45, sd=5) # high concentration at the right
d2 <- rnorm(n=200, mean=30, sd=10)
d3 <- rnorm(n=100, mean=15, sd=15) # flat tail at the left

# combine datasets
d123 <- c(d1, d2, d3)

plot_distribution(d123, "Distribution 2")
```

Code 1. Code for Distribution 2 (Q1a).

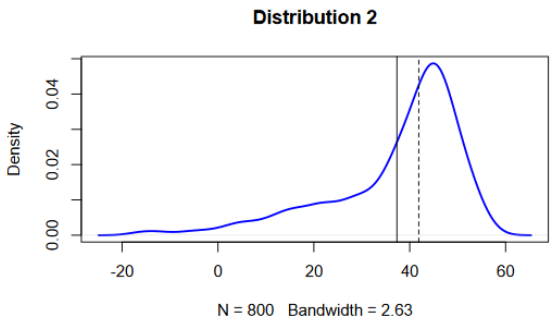


Figure 1. Plot for Distribution 2 (Q1a)

(b) Distribution 3: a single dataset that is normally distributed (bell-shaped, symmetric) using the rnorm() function to create a single large dataset (n=800).

```
# Single normally distributed dataset
d_symmetric <- rnorm(n=800, mean=30, sd=10)

plot_distribution(d_symmetric, "Distribution 3")
```

Code 2. Code for Distribution 3 (Q1b).

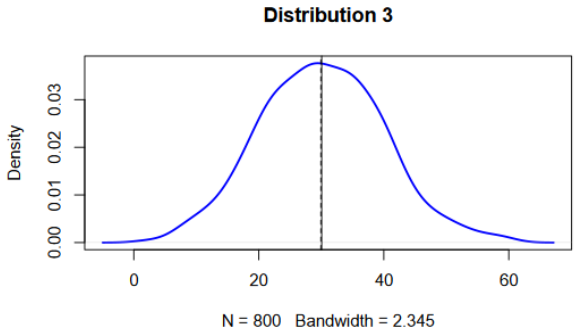


Figure 2. Plot for Distribution 3 (Q1b)

(c) Which measure of central tendency (mean or median) will be more sensitive (will change more) to outliers being added to your data?

- The mean will be more sensitive to outliers, as the mean is calculated by finding the sum for all values. Adding an outlier will significantly change the total, while the median only looks at the middle position, making it a more robust measure of central tendency.

Reusable function that does the plot for Q2:

```
plot_distribution2 <- function(data, title) {
  # plot the density function of data
  plot(density(data), col="blue", lwd=2, main=title)

  # calculate mean and standard deviation
  m <- mean(data)
  s <- sd(data)

  # for mean (solid)
  abline(v = m)

  # 1st, 2nd, and 3rd standard deviations left and right (dashed)
  abline(v = c(m-s, m-2*s, m-3*s, m+s, m+2*s, m+3*s), lty="dashed")
}
```

Code 3. Reusable function for Q2's plot.

Q2.(a) A random dataset (rdata) that is normally distributed with: n=2000, mean=0, sd=1, with a solid vertical line on the mean, and dashed vertical lines at the 1st, 2nd, and 3rd standard deviations to the left and right of the mean.

```
rdata <- rnorm(n=2000, mean=0, sd=1)
plot_distribution2(rdata, "Standard Normal Distribution (Q2A)")
```

Code 4. Code for Q2a.

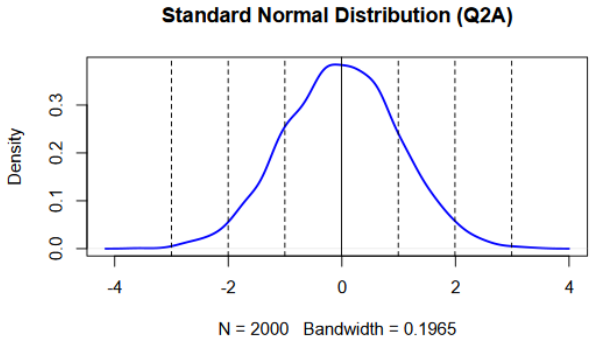


Figure 3. Plot for Q2a.

(b) Data points that correspond to the 1st, 2nd, and 3rd quartiles of rdata, and how many standard deviations away from the mean those points are corresponding to the 1st, 2nd, and 3rd quartiles.

```
# find the 1st, 2nd, 3rd quartiles
q_rdata <- quantile(rdata, c(0.25, 0.50, 0.75))

# calculate SDs away from mean (Z-scores)
z_rdata <- (q_rdata - mean(rdata)) / sd(rdata)

print(q_rdata)
print(z_rdata)
```

Code 5. Code for Q2b.

Data point value:

25% (1st quartile)	50% (2nd quartile)	75% (3rd quartile)
-0.688588497982307	0.0038215988929416016	0.6627989350133598

Z-scores:

25% (1st quartile)	50% (2nd quartile)	75% (3rd quartile)
-0.6846554734904474	0.008896668328053524	0.6689609061817517

(c) A new random dataset that is normally distributed with: n=2000, mean=35, sd=3.5, then determine how many standard deviations from the mean correspond to the 1st and 3rd quartiles and compare the result with (b).

```
# create the new random dataset
rdata_new <- rnorm(n=2000, mean=35, sd=3.5)

# find the 1st, 3rd quartiles
q_new <- quantile(rdata_new, c(0.25, 0.75))

# calculate SDs away from mean (Z-scores)
z_new <- (q_new - mean(rdata_new)) / sd(rdata_new)

print(z_new)
```

Code 6. Code for Q2c.

Z-Scores:

25% (1st quartile)	75% (3rd quartile)
-0.6457948240561594	0.672606886398341

Comparison with 2(b):
The number of standard deviations away from the mean for the quartiles in this distribution is very similar to those in rdata of 2(b). This is because both datasets are normally distributed. For any normal distribution, the 1st and 3rd quartiles always maintain the same relative distance from the mean, regardless of the specific values for the mean or standard deviation.

We can see that results for both (b) and (c) show that the 1st and 3rd quartiles are at approximately 0.67 standard deviations away (below for 1st quartile, and above for 3rd quartile) from the mean.

(d) For the dataset d123 from Question 1, determine how many standard deviations from the mean correspond to the 1st and 3rd quartiles and compare the result with (b)

```
q_d123 <- quantile(d123, c(0.25, 0.75))
z_d123 <- (q_d123 - mean(d123)) / sd(d123)

print(z_d123)
```

Code 7. Code for Q2d.

Z-Scores:

25% (1st quartile)	75% (3rd quartile)
-0.4055613846224141	0.6686673973472447

Comparison with 2(b):
Since d123 is negatively skewed, the mean is pulled down toward the long left tail. This makes the distance between the mean and the 1st quartile much smaller (-0.406) compared to the normal distribution's (-0.685). However, the Q3 Z-score in both distributions is similar (approx. 0.67), which means that the right-hand part of the d123 distribution still behaves like a normal curve, while the left-hand side is where the skewness occurs.

Q3. (a) Which formula does Rob Hyndman's answer (1st answer) suggest using for bin widths/number? Also, what does the Wikipedia article say is the benefit of that formula?

- The formula suggested to be used is $h = 2 \times IQR \times n^{-1/3}$, where n is the number of observations.
- According to Wikipedia, the benefit of this formula is that it is less sensitive to outliers than other rules (like Scott's rule) because it uses the interquartile range rather than the standard deviation.

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(b) Given a random normal distribution rand_data <- rnorm(800, mean=20, sd = 5), compute the bin widths (h) and number of bins (k) according to Sturges' formula, Scott's normal reference rule (uses standard deviation), and Freedman-Diaconis' choice (uses IQR)

```
# generate random normal distribution
n <- 800

rand_data <- rnorm(n, mean=20, sd=5)

# range for k/h conversions
range <- max(rand_data) - min(rand_data)

# Sturges' Formula
k_sturges <- ceiling(log2(n) + 1)

h_sturges <- range / k_sturges

print(k_sturges)
[1] 11

print(h_sturges)
[1] 2.996035

# Scott's Normal Reference Rule
h_scott <- (3.49 * sd(rand_data)) / (n^(1/3))

k_scott <- ceiling(range / h_scott)

print(h_scott)
[1] 1.931376

print(k_scott)
[1] 18

# Freedman-Diaconis' Choice
h_fd <- (2 * IQR(rand_data)) / (n^(1/3))

k_fd <- ceiling(range / h_fd)

print(h_fd)
[1] 1.449131

print(k_fd)
[1] 23
```

Code 8. Code with Output Values for Q3b.

(c) Repeat part (b) with out_data <- c(rand_data, runif(10, min=40, max=60)). Which of the three methods does the bin width (h) change the least when outliers are added?

```
out_data <- c(rand_data, runif(10, min=40, max=60))

range2 <- max(out_data) - min(out_data)

n2 <- length(out_data)

# Sturges' Formula
k_sturges2 <- ceiling(log2(n2) + 1)

h_sturges2 <- range2 / k_sturges2

print(k_sturges2)
[1] 11

print(h_sturges2)
[1] 5.047831

# Scott's Normal Reference Rule
h_scott2 <- (3.49 * sd(out_data)) / (n^(1/3))

k_scott2 <- ceiling(range2 / h_scott2)

print(h_scott2)
[1] 2.255446

print(k_scott2)
[1] 25

# Freedman-Diaconis' Choice
h_fd2 <- (2 * IQR(out_data)) / (n2^(1/3))

k_fd2 <- ceiling(range2 / h_fd2)

print(h_fd2)
[1] 1.461478

print(k_fd2)
[1] 38
```

Code 9. Code with Output Values for Q3c.

Method where the bin width (h) change the least:

Sturges' Formula	|2.996035 - 5.047831| = 2.051796
Scott's Normal Reference Rule	|1.931376 - 2.255446| = 0.32407
Freedman-Diaconis' Choice	|1.449131 - 1.461478| = 0.012347

This is consistent with the answer at 3(a). This is because the Interquartile Range (IQR) only looks at the middle 50% of the data (the distance between the 1st and 3rd quartiles). Because outliers live at the extreme 1% or 0.1% of the dataset, they do not move the 1st or 3rd quartiles, making it less sensitive to outliers.